Emergent Clathrin Coat Stiffening Explains Different Pathways

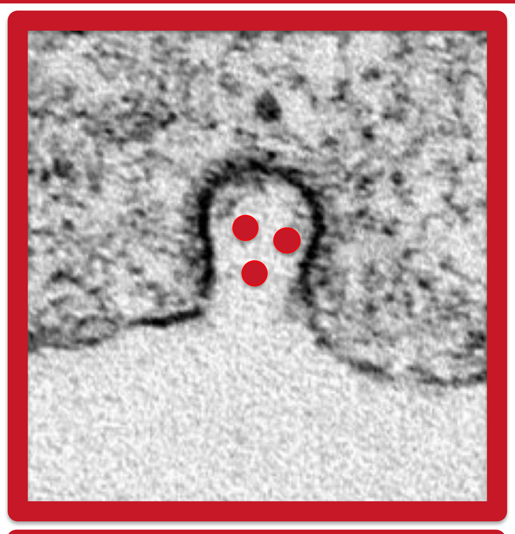
Johannes Dreckhoff,
Leon Lettermann, Ulrich Schwarz

Institute for Theoretical Physics, Heidelberg University, Germany
BioQuant, Heidelberg University, Germany

Get in touch: **j.dreckhoff@gmail.com**

Motivation

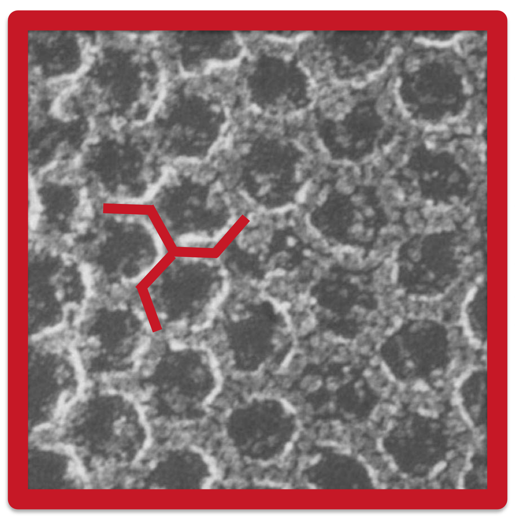
During certain transport processes of cargo across membranes (endocytosis), the **membrane bends** around the cargo (red dots). To overcome the energy barrier, the cell assembles a protein coat (black).
One of the four proteins doing so is called clathrin. It is remarkable because of its striking geometry out of three **clockwise bent legs**, which attach in the same chiral orientation onto the membrane. To each leg, two other clathrin cores can bind.


Snapshot during endocytosis [1]


Electron micrograph of clathrin [2]

This enables clathrin to build **highly regular hexagonal lattices**. Since the structural scale of the grid is below the Bragg limit, its temporal evolution is not yet experimentally accessible. This leaves the following questions open:

- Why is the coated membrane (~300 k_BT) so much stiffer than the membrane and clathrin (~20 k_BT) individually?
- When/How are defects (pentagons) included, are they necessary to generate curvature?
- What is the relation of coat area and coat curvature during growth?


Hexagonal clathrin lattice with triskelion [3]

Scan me to find more details about the CME, current models and our approach to simulate structural assembly and curvature generation during CME:

We created a modular software package called `endopythosis` for simulating the **structural kinetic assembly** of clathrin on the **deformable membrane**. It uses a simple and interpretable representation of clathrin and takes into account the full **Helfrich**, **binding** and **deformation energies**. The membrane can **dynamically change** its **curvature** during evolution, granting full access to the **time evolution** of e.g. defect number and distribution, internal connectivity and effective bending rigidity.

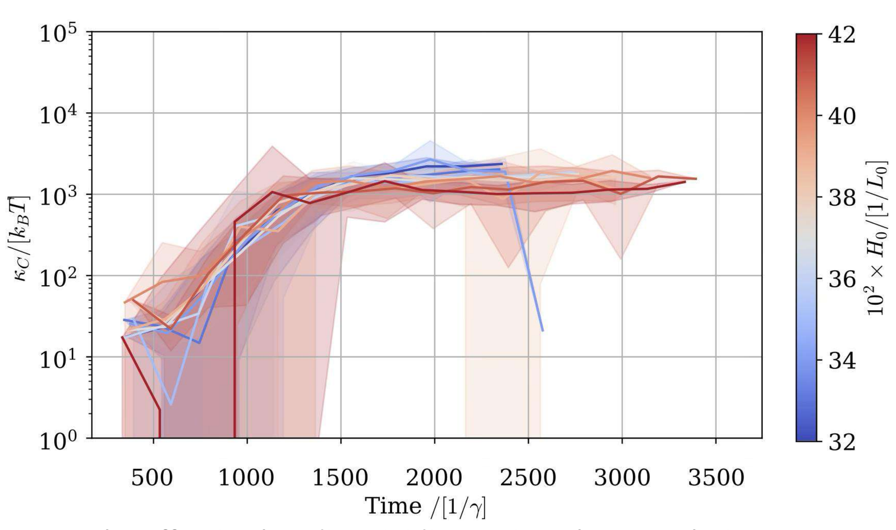
endopythosis
STRUCTURAL SIMULATION LIBRARY

Scan me to find our repository, where we will soon make public the package code:

Clathrin Stiffens up

An **effective stiffening** of the clathrin-membrane complex emerges in our simulations. This is due to a shift in **how deformations are mediated**.

- Small coats: deformation of bond joints (~20 k_BT)
- Large coats: deformation of bond lengths (~ 800 k_BT)

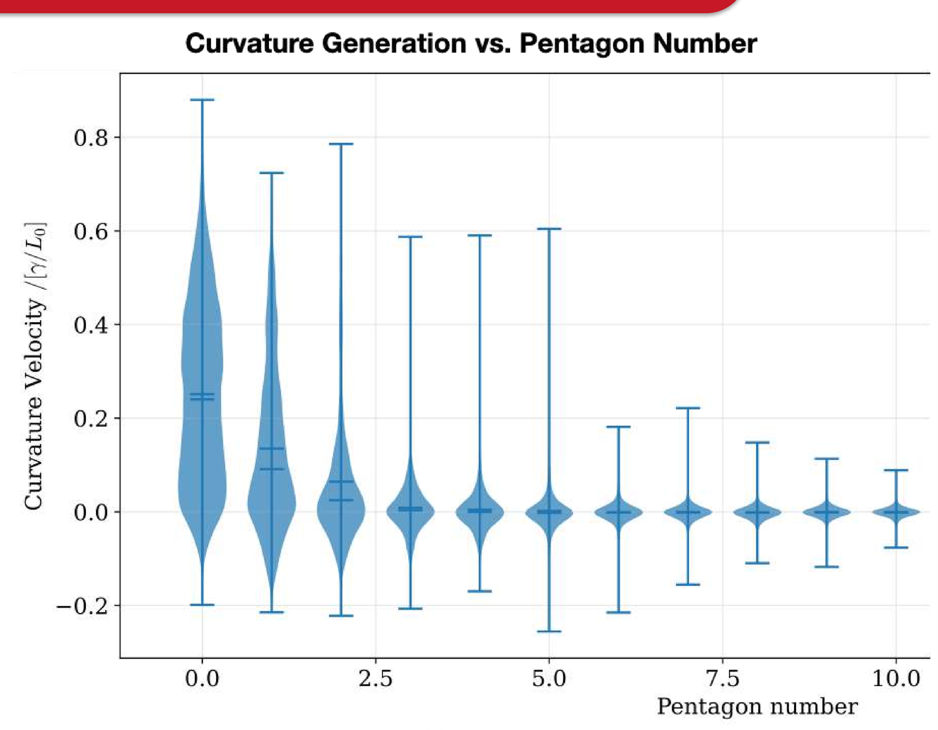


The effective bending rigidity against the growth time.
An increase of ~2 orders of magnitude can be observed.

This can be described by two springs in series, where the weaker one stiffens up.

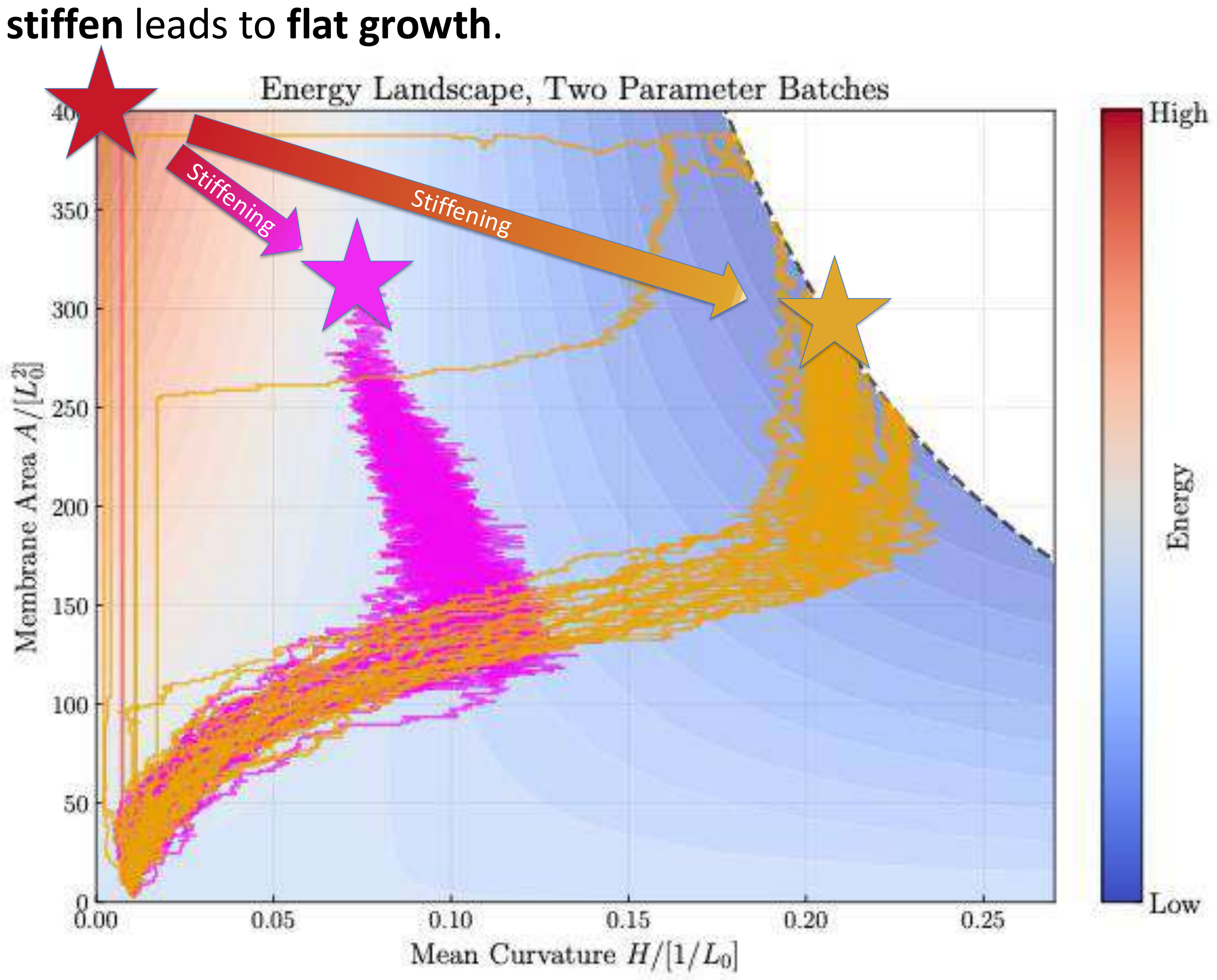
Defects do not Drive Curvature Generation

Most of the curvature is generated **without** any **defects present**, leading us to believe that defects are **not causing** curvature generation, but are a **symptom of** (and can "lock-in") curvature.

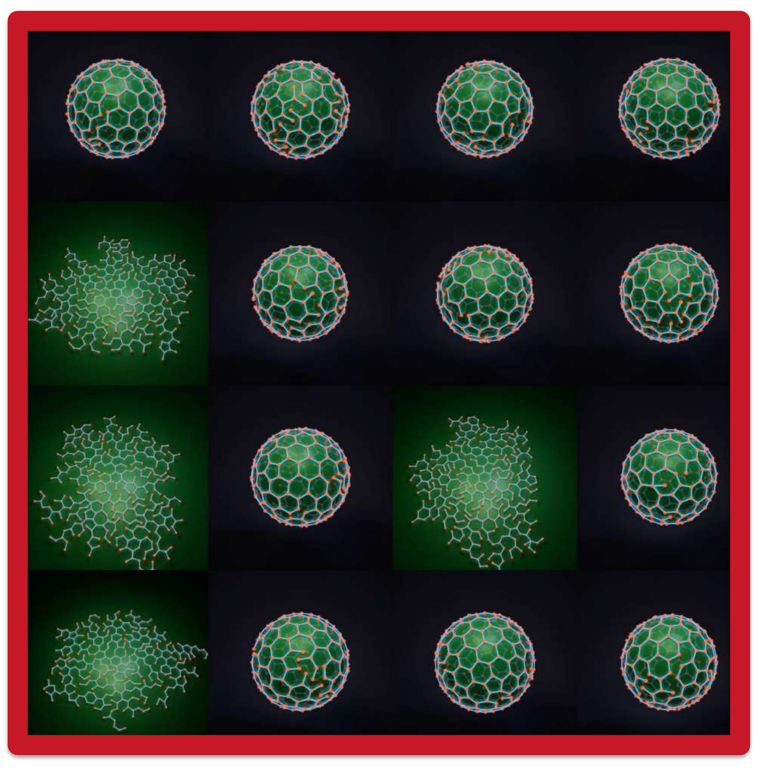


Stiffening Shifts the Energy Landscape

Three main energy contributions shape the energy landscape: membrane energy, polymerisation energy and clathrin bending energy. **Stiffening** of clathrin increases the contribution of the latter, **shifting the energy basin** (stars) towards higher curvatures. Evolution stops as the state approaches the basin. **Failure to stiffen** leads to **flat growth**.



Different Curvature Fates Emerge



Under equal conditions, systems develop **different "curvature fates"**, ending either flat, partially invaginated (aborted) or fully closed

Scan me to find additional material like plots or movies for an easier understanding of the time evolution:

1: Volker et al., Membrane Remodeling in Clathrin-Mediated Endocytosis, JCB 2018, 2: Bruce Alberts, Molecular Biology of the Cell 2022, 3: J. Heuser, 3D visualisation of coated vesicle formation in fibroblasts, JCB 1980